

Experiment ‘p_1_10_animal_speaking’ Results

June 4, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 2

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 2

Problem Specification

Problem name: p_1_10_animal_speaking

Natural language statement: null

Method signature: p_1_10_animal_speaking() returns (s: string)

Ensures

- `s == " /__/__ /---\\\n (' ') / Hello \\\n (-) | New |\n | | | __ Coder /\n (__|__)
\\-----/\n"`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

`p_1_10_animal_speaking()` returns (`s: string`)

The method should respect the following contract:

`ensures s == " /\_/_/\ /---\\n ( ' ' ) / Hello \\n ( - ) | New |\n | | | \\ Coders /\n (`
→ `__|__ ) \\----/\n"`

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn2IzX9jDlpf4Hf5bSgOqV7tKceAW

Model: gpt-5.5-2026-04-23

Created at: 1780579305

Finish reason: stop

Usage: 342 tokens in, and 4261 tokens out

```
{
  s := " /\_/_/\ /---\\" + "\n" +
    " ( ' ' ) / Hello \\" + "\n" +
    " ( - ) | New |" + "\n" +
    " | | | \\ Coders /\n" +
    " (__|__) \\----/" + "\n";
}
```

Program → GenAI

When using dafny verify, the below error is emitted and verify fails:

problem_data/p_1_10_animal_speaking/program.dfy(3,0): Error: a postcondition could not be
→ proved on this return path
|
3 | {
| ^

```

problem_data/p_1_10_animal_speaking/program.dfy(2,11): Related location: this is the
    ↪ postcondition that could not be proved
|
2 | ensures s == " /\_\_/\_\_ /---\\n ( ' ' ) / Hello \\n ( - ) | New |n | | | \\ Coders
    ↪ /\n ( __|__ ) \\----/\n"
| ^^

Dafny program verifier finished with 0 verified, 1 error

Correct the error in code by altering only the method body. Produce and show only the
    ↪ Dafny body, including the curly braces that surround it. Do not show the signature
    ↪ nor contract. You must not use 'assume' nor 'expect' anywhere in your code.

```

GenAI → Program

System fingerprint: null
ID: chatcmpl-Dn2L4HXMSWvgIN72Go5hHMBIBD4qa
Model: gpt-5.5-2026-04-23
Created at: 1780579434
Finish reason: stop
Usage: 773 tokens in, and 3706 tokens out

```

{
  s := " /\_\_/\_\_ /---\\n ( ' ' ) / Hello \\n ( - ) | New |n | | | \\ Coders /\n ( __|__ )
    ↪ \\----/\n";
}

```

Final Program

```

method p_1_10_animal_speaking() returns (s: string)
  ensures s == " /\_\_/\_\_ /---\\n ( ' ' ) / Hello \\n ( - ) | New |n | | | \\
    ↪ Coders /\n ( __|__ ) \\----/\n"
{
  s := " /\_\_/\_\_ /---\\n ( ' ' ) / Hello \\n ( - ) | New |n | | | \\ Coders /\n ( __|__ )
    ↪ \\----/\n";
}

```

Total Token Usage

Input tokens: 1115
Output tokens: 7967
Reasoning tokens: 7676
Sum of ‘total tokens’: 9082

Experiment Timings

Overall Experiment started at 1780579304333, ended at 1780579553735, lasting 249402ms (249.40 seconds)
Iteration #1 started at 1780579304334, ended at 1780579432309, lasting 127975ms (127.98 seconds)

Iteration #2 started at 1780579432312, ended at 1780579553734, lasting 121422ms (121.42 seconds)

Experiment 'p_1_1_greeting' Results

June 4, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_1_1_greeting

Natural language statement: null

Method signature: p_1_1_greeting() returns (s: string)

Ensures

- s == "Hugo"

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

`p_1_1_greeting()` returns (`s: string`)

The method should respect the following contract:

`ensures s == "Hugo"`

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn2BHf1HGwhU3M8N1ksOVlnXy057y

Model: gpt-5.5-2026-04-23

Created at: 1780578827

Finish reason: stop

Usage: 238 tokens in, and 23 tokens out

```
{  
  s := "Hugo";  
}
```

Final Program

```
method p_1_1_greeting() returns (s: string)  
  ensures s == "Hugo"  
{  
  s := "Hugo";  
}
```

Total Token Usage

Input tokens: 238

Output tokens: 23

Reasoning tokens: 0

Sum of ‘total tokens’: 261

Experiment Timings

Overall Experiment started at 1780578818414, ended at 1780578864248, lasting 45834ms (45.83 seconds)

Iteration #1 started at 1780578818445, ended at 1780578864247, lasting 45802ms (45.80 seconds)

Experiment 'p_1_2_sum' Results

June 4, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_1_2_sum

Natural language statement: null

Method signature: p_1_2_sum() returns (i: int)

Ensures

- $i == (10 * 11) / 2$

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

`p_1_2_sum()` returns `(i: int)`

The method should respect the following contract:

`ensures i == (10 * 11) / 2`

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn2Budp2ASG41TQ2cFGTpRJWoG87I

Model: gpt-5.5-2026-04-23

Created at: 1780578866

Finish reason: stop

Usage: 248 tokens in, and 76 tokens out

```
{  
  i := (10 * 11) / 2;  
}
```

Final Program

```
method p_1_2_sum() returns (i: int)  
  ensures i == (10 * 11) / 2  
{  
  i := (10 * 11) / 2;  
}
```

Total Token Usage

Input tokens: 248

Output tokens: 76

Reasoning tokens: 42

Sum of ‘total tokens’: 324

Experiment Timings

Overall Experiment started at 1780578865732, ended at 1780578904714, lasting 38982ms (38.98 seconds)

Iteration #1 started at 1780578865733, ended at 1780578904714, lasting 38981ms (38.98 seconds)

Experiment ‘p_1_3_product’ Results

June 4, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_1_3_product

Natural language statement: null

Method signature: p_1_3_product() returns (i: int)

Ensures

- `i == factorial(10)`

Functional Code Given

```
function factorial(n: int): int
  decreases n
{
  if n <= 1 then 1 else n * factorial(n - 1)
}
```

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

`p_1_3_product()` returns (`i: int`)

The method should respect the following contract:

`ensures i == factorial(10)`

The contract uses the following dafny code:

```
function factorial(n: int): int
  decreases n
{
  if n <= 1 then 1 else n * factorial(n - 1)
}
```

These function(s) must not be used in your implementation of the method.

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn2CgWv91etbH7B3x06XE7AgmWNaA

Model: gpt-5.5-2026-04-23

Created at: 1780578914

Finish reason: stop

Usage: 348 tokens in, and 541 tokens out

```
{
  i := 3628800;
}
```

Final Program

```
function factorial(n: int): int
  decreases n
{
```

```
    if n <= 1 then 1 else n * factorial(n - 1)
  }

method p_1_3_product() returns (i: int)
  ensures i == factorial(10)
{
  i := 3628800;
}
```

Total Token Usage

Input tokens: 348

Output tokens: 541

Reasoning tokens: 512

Sum of 'total tokens': 889

Experiment Timings

Overall Experiment started at 1780578910348, ended at 1780578967792, lasting 57444ms (57.44 seconds)

Iteration #1 started at 1780578910521, ended at 1780578967791, lasting 57270ms (57.27 seconds)

Experiment ‘p_1_4_account_balance’ Results

June 4, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_1_4_account_balance

Natural language statement: null

Method signature: p_1_4_account_balance() returns (first: real, second: real, third: real)

Ensures

- `first == 1000.0 + 1000.0 * 0.05`
- `second == first + first * 0.05`
- `third == second + second * 0.05`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

`p_1_4_account_balance()` returns (first: real, second: real, third: real)

The method should respect the following contract:

ensures first == 1000.0 + 1000.0 * 0.05, ensures second == first + first * 0.05, ensures
 ↪ third == second + second * 0.05

Produce and show only the Dafny body of this method, including the curly braces that
 ↪ surround it. Do not show the signature nor contract. You must not use 'assume' nor
 ↪ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn2DaIcCkQuoyuMEiUfhndvdG1w3w

Model: gpt-5.5-2026-04-23

Created at: 1780578970

Finish reason: stop

Usage: 328 tokens in, and 185 tokens out

```
{
  first := 1000.0 + 1000.0 * 0.05;
  second := first + first * 0.05;
  third := second + second * 0.05;
}
```

Final Program

```
method p_1_4_account_balance() returns (first: real, second: real, third: real)
  ensures first == 1000.0 + 1000.0 * 0.05
  ensures second == first + first * 0.05
  ensures third == second + second * 0.05
{
  first := 1000.0 + 1000.0 * 0.05;
  second := first + first * 0.05;
  third := second + second * 0.05;
}
```

Total Token Usage

Input tokens: 328

Output tokens: 185

Reasoning tokens: 118

Sum of 'total tokens': 513

Experiment Timings

Overall Experiment started at 1780578969282, ended at 1780579010078, lasting 40796ms (40.80 seconds)

Iteration #1 started at 1780578969283, ended at 1780579010078, lasting 40795ms (40.80 seconds)

Experiment ‘p_1_5_box’ Results

June 4, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_1_5_box

Natural language statement: null

Method signature: p_1_5_box() returns (s: string)

Ensures

- `s == "+-----+\n| Dave |\n+-----+\n"`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

`p_1_5_box()` returns `(s: string)`

The method should respect the following contract:

`ensures s == "+-----+\n| Dave |\n+-----+\n"`

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn2EKzlo7NJoDNsi779K0tYgZNUqN

Model: gpt-5.5-2026-04-23

Created at: 1780579016

Finish reason: stop

Usage: 254 tokens in, and 96 tokens out

```
{  
  s := "+-----+\n| Dave |\n+-----+\n";  
}
```

Final Program

```
method p_1_5_box() returns (s: string)  
  ensures s == "+-----+\n| Dave |\n+-----+\n"  
{  
  s := "+-----+\n| Dave |\n+-----+\n";  
}
```

Total Token Usage

Input tokens: 254

Output tokens: 96

Reasoning tokens: 56

Sum of ‘total tokens’: 350

Experiment Timings

Overall Experiment started at 1780579014447, ended at 1780579087271, lasting 72824ms (72.82 seconds)

Iteration #1 started at 1780579014489, ended at 1780579087267, lasting 72778ms (72.78 seconds)

Experiment ‘p_1_6_large_letters’ Results

June 4, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_1_6_large_letters

Natural language statement: null

Method signature: p_1_6_large_letters() returns (s: string)

Ensures

- `s == "* * * ***** * *\n* * * * * * * *\n***** * * ***** * *\n* * * * * * * *"`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

p_1_6_large_letters() returns (s: string)

The method should respect the following contract:

```
ensures s == "* * * ***** ***** * *\n* * * * * * * * *\n***** * * ***** ***** * *\n* *  
↪ ***** * * * * *\n* * * * * * * * *"
```

Produce and show only the Dafny body of this method, including the curly braces that
↪ surround it. Do not show the signature nor contract. You must not use 'assume' nor
↪ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn2Fn73M1IdOOTikFUUNkpJpyreKr

Model: gpt-5.5-2026-04-23

Created at: 1780579107

Finish reason: stop

Usage: 400 tokens in, and 204 tokens out

```
{  
  s := "* * * ***** ***** * *\n* * * * * * * * *\n***** * * ***** ***** * *\n* *  
↪ ***** * * * * *\n* * * * * * * * *";  
}
```

Final Program

```
method p_1_6_large_letters() returns (s: string)  
  ensures s == "* * * ***** ***** * *\n* * * * * * * * *\n***** * * ***** ***** * *\n* *  
↪ ***** * * * * *\n* * * * * * * * *"  
{  
  s := "* * * ***** ***** * *\n* * * * * * * * *\n***** * * ***** ***** * *\n* *  
↪ ***** * * * * *\n* * * * * * * * *";  
}
```

Total Token Usage

Input tokens: 400

Output tokens: 204

Reasoning tokens: 93

Sum of 'total tokens': 604

Experiment Timings

Overall Experiment started at 1780579105759, ended at 1780579157555, lasting 51796ms (51.80 seconds)

Iteration #1 started at 1780579105788, ended at 1780579157554, lasting 51766ms (51.77 seconds)

Experiment ‘p_1_7_face’ Results

June 4, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_1_7_face

Natural language statement: null

Method signature: p_1_7_face() returns (s: string)

Ensures

- s == " ////\n +\"\"\"\"\"\"+\"n(| o o |)\n | ^ |\n | ‘-’ |\n +-----+\n"

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

`p_1_7_face()` returns (`s: string`)

The method should respect the following contract:

`ensures s == " /////\n +\"\\\"\\\"\\\"\"+\\n(| o o |)\n | ^ |\n | ‘’- |\n +-----+\n"`

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn2GgDjewadhpFuXhXEfTyHlOzWq7

Model: gpt-5.5-2026-04-23

Created at: 1780579162

Finish reason: stop

Usage: 300 tokens in, and 443 tokens out

```
{  
  s := " /////\n +\"\\\"\\\"\\\"\\\"\"+\\n(| o o |)\n | ^ |\n | ‘’- |\n +-----+\n";  
}
```

Final Program

```
method p_1_7_face() returns (s: string)  
  ensures s == " /////\n +\"\\\"\\\"\\\"\\\"\"+\\n(| o o |)\n | ^ |\n | ‘’- |\n +-----+\n"  
{  
  s := " /////\n +\"\\\"\\\"\\\"\\\"\"+\\n(| o o |)\n | ^ |\n | ‘’- |\n +-----+\n";  
}
```

Total Token Usage

Input tokens: 300

Output tokens: 443

Reasoning tokens: 372

Sum of ‘total tokens’: 743

Experiment Timings

Overall Experiment started at 1780579162148, ended at 1780579202189, lasting 40041ms (40.04 seconds)

Iteration #1 started at 1780579162217, ended at 1780579202189, lasting 39972ms (39.97 seconds)

Experiment ‘p_1_8_painting’ Results

June 4, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_1_8_painting

Natural language statement: null

Method signature: p_1_8_painting() returns (s: string)

Ensures

- `s == mystring(" !!!@@@@@@@@@@@@@@@@@@@@\n", 5)`

Functional Code Given

```
function mystring(s: string, n: nat): string
{
  if n == 0 then "" else s + mystring(s, n - 1)
}
```

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

`p_1_8_painting()` returns `(s: string)`

The method should respect the following contract:

`ensures s == mystring(" !!!@@@@@@@@@@@@@@@@@@@@\n", 5)`

The contract uses the following dafny code:

```
function mystring(s: string, n: nat): string
{
  if n == 0 then "" else s + mystring(s, n - 1)
}
```

These function(s) must not be used in your implementation of the method.

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn2HPySD71FNMvIBsfM6ZOgNXB8oV

Model: gpt-5.5-2026-04-23

Created at: 1780579207

Finish reason: stop

Usage: 378 tokens in, and 493 tokens out

```
{
  s := " !!!@@@@@@@@@@@@@@@@@@@@\n" +
    (" !!!@@@@@@@@@@@@@@@@@@@@\n" +
    (" !!!@@@@@@@@@@@@@@@@@@@@\n" +
    (" !!!@@@@@@@@@@@@@@@@@@@@\n" +
    (" !!!@@@@@@@@@@@@@@@@@@@@\n" + ""))));
}
```

Final Program

```

function mystring(s: string, n: nat): string
{
  if n == 0 then "" else s + mystring(s, n - 1)
}

method p_1_8_painting() returns (s: string)
  ensures s == mystring(" !!!@@@@@@@@@@@@@@@@@@@@@\\n", 5)
{
  s := " !!!@@@@@@@@@@@@@@@@@@@@@\\n" +
    (" !!!@@@@@@@@@@@@@@@@@@@@@\\n" +
    (" !!!@@@@@@@@@@@@@@@@@@@@@\\n" +
    (" !!!@@@@@@@@@@@@@@@@@@@@@\\n" +
    (" !!!@@@@@@@@@@@@@@@@@@@@@\\n" + "")))));
}

```

Total Token Usage

Input tokens: 378

Output tokens: 493

Reasoning tokens: 398

Sum of ‘total tokens’: 871

Experiment Timings

Overall Experiment started at 1780579206875, ended at 1780579265999, lasting 59124ms (59.12 seconds)

Iteration #1 started at 1780579206883, ended at 1780579265999, lasting 59116ms (59.12 seconds)

Experiment ‘p_1_9_house’ Results

June 4, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_1_9_house

Natural language statement: null

Method signature: p_1_9_house() returns (s: string)

Ensures

- `s == " +\n + +\n + +\n+-----+\n| .-. | \n| | | \n+---+\n"`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

`p_1_9_house()` returns (`s: string`)

The method should respect the following contract:

`ensures s == " +\n + +\n + +\n+-----+\n| .-. |\n| | | |\n+--+--++\n"`

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn2IPmvrX5FzJ3cvjbnU4Rlu41DEE

Model: gpt-5.5-2026-04-23

Created at: 1780579269

Finish reason: stop

Usage: 296 tokens in, and 57 tokens out

```
{  
  s := " +\n + +\n + +\n+-----+\n| .-. |\n| | | |\n+--+--++\n";  
}
```

Final Program

```
method p_1_9_house() returns (s: string)  
  ensures s == " +\n + +\n + +\n+-----+\n| .-. |\n| | | |\n+--+--++\n"  
{  
  s := " +\n + +\n + +\n+-----+\n| .-. |\n| | | |\n+--+--++\n";  
}
```

Total Token Usage

Input tokens: 296

Output tokens: 57

Reasoning tokens: 0

Sum of ‘total tokens’: 353

Experiment Timings

Overall Experiment started at 1780579267850, ended at 1780579299707, lasting 31857ms (31.86 seconds)

Iteration #1 started at 1780579267854, ended at 1780579299707, lasting 31853ms (31.85 seconds)

